

Markscheme

Mathematics: analysis and approaches

Practice question bank

153. (a)

Given that $z = re^{i\theta}$, where $r > 0$.

Simplify $\frac{z^3}{\bar{z}}$ in terms of r and θ , and hence evaluate it when $r = 2$ and $\theta = \frac{\pi}{4}$, giving the answer in the form $a + bi$.

$$\bar{z} = re^{-i\theta}, \text{ so } \frac{z^3}{\bar{z}} = \frac{r^3 e^{3i\theta}}{r e^{-i\theta}} = r^2 e^{4i\theta} \quad (M1) \text{ A1}$$

$$\text{substitute } r = 2, \theta = \frac{\pi}{4}: 4e^{i\pi} \quad (M1)$$

$$= 4(\cos 180^\circ + i \sin 180^\circ) \quad (M1)$$

$$= -4 \quad \text{A1}$$

[6 marks]